Product Requirements Document (PRD)

Product: SUEZ AI Gateway

Version: 1.0

Date: March 2026

1. Purpose

SUEZ is an AI control gateway that sits between applications and AI systems including LLMs, MCP tools, and AI agents. All AI traffic must flow through SUEZ.

2. Problem Statement

- No centralized policy enforcement
- No unified audit trail
- Vendor lock-in
- Uncontrolled tool execution
- Opaque agent behavior
- Cost overruns from unmanaged token usage

3. Goals

- Single enforcement point for AI workloads
- Multi-provider model support
- Central tool governance
- Agent behavior constraints
- Full request traceability
- Quota and rate enforcement
- Horizontal scalability

4. Policy Engine

Default engine: Open Policy Agent (OPA).

Support pluggable custom policy engine interface.

- Rego-based policies
- Versioned policies in control plane
- Logged policy inputs and decisions
- Stateless evaluation

5. Token Governance

Default token limit equals maximum allowed by selected LLM provider.

- Provider metadata awareness
- Policy override support
- Hard enforcement at provider max
- Token usage logging

6. Environment Isolation

Isolation enforced at database level.

- Separate databases per environment (dev, staging, prod)
- Separate audit storage per environment
- No cross-environment policy access
- Environment-scoped RBAC enforcement

7. Security Requirements

- TLS encryption in transit
- Secure vault-based secret storage
- Secret rotation support
- Role-based authorization

- Immutable audit logging

8. Non-Functional Requirements

- Latency overhead < 150ms
- Availability >= 99.9%
- Horizontal scalability
- Tamper-evident audit logs
- Database-level environment isolation

9. Summary

SUEZ provides centralized governance for models, tools, and agents. It enforces identity, policy, auditability, and operational discipline across enterprise AI systems.